

Abeer Banerjee

PhD Candidate

Computational Imaging

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About

PhD researcher in computational imaging, specializing in physics-informed learning for solving inverse problems. Experience in learned reconstruction for lensless imaging and optoacoustic microscopy under low-data constraints. Published in leading venues (BMVC, ICIP, IJCNN, IEEE Transactions, Elsevier journals). Currently at TranslaTUM-TUM Center for Translational Cancer Research.

Research Experience

DAAD Bi-national PhD Fellow

Oct 2024 – Present

Technical University of Munich (TUM), Germany

- Designing neural model-based reconstruction pipelines for optoacoustic microscopy using system impulse response (TIR) modeling
- Developing physics-informed deep learning approaches for improved axial resolution and physically consistent reconstruction
- Research conducted at the Institute of Biological and Medical Imaging (IBMI)

Senior Research Fellow

Sep 2022 – Sep 2025

CSIR-Central Electronics Engineering Research Institute (CEERI), India

- Developed physics-informed deep learning frameworks for lensless image reconstruction in low-data regimes
- Proposed learning-based reconstruction methods under sparse and noisy measurements
- Built AI models for eye-gaze estimation in extreme low-light using neuromorphic imaging
- Mentored junior researchers and contributed to scientific writing and peer review

Junior Research Fellow

Sep 2020 – Sep 2022

CSIR-CEERI, India

- Developed GAN-based frameworks for grayscale image restoration and colorization
- Contributed to deep learning models for face anti-spoofing in unconstrained environments

UG Research Intern

Jun 2018 – Aug 2018

Defence Research and Development Organization (DRDO), Govt. of India

- Implemented edge-preserving smoothing algorithms for infrared thermal imaging at IRDE DehraDun

Education

• Integrated PhD – Computational Imaging

2020 – Present

Academy of Scientific and Innovative Research (AcSIR), India

Advisors: Dr. Sanjay Singh (AcSIR), Prof. Vasilis Ntziachristos (TUM)

Master's Thesis (2022): Generative Colorization of Grayscale Images

Published as a journal paper: *ParaColorizer*, The Visual Computer, Springer, 2023, [Link]

CGPA: 9.13 / 10

• B.Tech – Electronics and Communication Engineering

2016 – 2020

University of Calcutta, India

Thesis Advisor: Prof. Ashik Paul

CGPA: 8.64 / 10

Publications (Selected)

Learned Reconstruction for Optoacoustic Microscopy

- Physics-Informed Neural Reconstruction in Optoacoustic Microscopy, A. Banerjee, L. Englert, D. Jüstel, S. Singh, and V. Ntziachristos, ISBI 2026, (Accepted Abstract)

Learned Lensless Image Reconstruction

- Towards Physics-informed Cyclic Adversarial Multi-PSF Lensless Imaging, **A. Banerjee**, and S. Singh, IEEE Transactions on ETCI, 2025, IF: 8.0. [Link].
- Towards Lensless Image Deblurring with Prior-Embedded INRs, **A. Banerjee**, and S. Singh, Engineering Applications of AI, Elsevier, 2025, IF: 8.0. [Link].
- Reconstructing Synthetic Lensless Images in Low-Data Regime, **A. Banerjee**, H. Kumar, S. Saurav, and S. Singh, BMVC 2023, h5-index: 77. [Link].
- Physics-informed Deep Deblurring, **A. Banerjee**, H. Kumar, S. Saurav, and S. Singh, ICIP 2023 (Oral), h5-index: 61. [Link].

Learned Neuromorphic Imaging for Eye-gaze estimation

- Generalized Gaze-Estimation in Low-light with Event-driven Neural Networks, **A. Banerjee**, N. K. Mehta, S. S. Prasad, H. Kumar, S. Saurav, and S. Singh, IJCNN 2024 (Oral), h5-index: 60. [Link].
- Gaze Estimation with Encoded Retinomorphc Events, **A. Banerjee**, S. S. Prasad, N. K. Mehta, H. Kumar, S. Saurav, and S. Singh IHCI 2022 (Oral), h5-index: 14. [Link].

Other Collaborative Research with Shared Contributions

- JS-SpoofNet: Jointly Supervised Neural Spoof Detection, Neurocomputing, 2023, IF: 6.5. [Link].
- Multimodal Dataset for Industrial Task Monitoring and Engagement Prediction, HRI 2025, h5-index: 51. [Link].

Skills

Research Proficiency

Inverse Problems, Computational Imaging, Image Reconstruction, Physics-informed Learning, Low-data Learning, Generative Modeling

AI

Deep Learning (PyTorch preferred), Neural Representations, Generative Models (GANs, implicit models, familiarity with diffusion models), Optimization for Inverse Problems

Programming and Tools

Python, MATLAB, OpenCV, Docker, TensorBoard, L^AT_EX

Voluntary Academic Service

- Session Chair, for session "Neural Networks for Image Processing" at IEEE IJCNN 2024
- Invited Reviewer: Neurocomputing, TVCJ, ESA, CEE, IET Computer Vision, and ranked conferences, e.g., IJCNN, ICVGIP, ACML, etc.

Awards and Achievements

- DAAD Bi-nationally Supervised PhD Fellowship (TUM, 2024-2026)
- GATE Fellowship 2020 (Junior Research Fellowship (2020-2023) & Senior Research Fellowship (2023-2025)), Government of India
- Fine Arts (Painting) 2013: 5th-year degree with distinction, Bangiya Sahitya Parishad, RBU.

Languages

English (Fluent), Bengali (Native), Hindi (Professional), German (Elementary)